

Linearization of an Affine Switched System

1 The affine switched system

Consider the affine system

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}, \quad x(t) \in \mathbb{R}^n. \quad (1)$$

The variable $t \in \mathbb{R}$ is continuous time. The switching signal $\sigma : \mathbb{R}_+ \rightarrow \mathcal{Q}$ is piecewise constant and selects one mode from the finite set $\mathcal{Q}$.

Let

$$t_0 < t_1 < \cdots < t_N$$

be the actual switching times, and let

$$\bar{t}_0 < \bar{t}_1 < \cdots < \bar{t}_N$$

be their nominal values. The prescribed mode on the i th interval is

$$q_i = \bar{\sigma}(\bar{t}_{i-1}).$$

Thus,

$$\sigma(t) = q_i = \bar{\sigma}(\bar{t}_{i-1}), \quad t_{i-1} \leq t < t_i, \quad i = 1, \dots, N. \quad (2)$$

Only the interval lengths are perturbed; the order of the modes remains fixed.

2 Conversion to a homogeneous system

The term $b_{\sigma(t)}$ makes (1) affine rather than homogeneous. To include this term in a matrix multiplication, augment the state with a constant component:

$$\mathcal{X}(t) = \begin{bmatrix} x(t) \\ 1 \end{bmatrix} \in \mathbb{R}^{n+1}.$$

Since the derivative of the last component is zero,

$$\dot{\mathcal{X}}(t) = \begin{bmatrix} \dot{x}(t) \\ 0 \end{bmatrix} = \begin{bmatrix} A_{\sigma(t)}x(t) + b_{\sigma(t)} \\ 0 \end{bmatrix}.$$

This can be written as

$$\dot{\mathcal{X}}(t) = \begin{bmatrix} A_{\sigma(t)} & b_{\sigma(t)} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x(t) \\ 1 \end{bmatrix}.$$

Define the augmented matrix associated with mode q by

$$\tilde{A}_q = \begin{bmatrix} A_q & b_q \\ 0 & 0 \end{bmatrix}.$$

The affine system is therefore equivalent to the homogeneous system

$$\boxed{\dot{\mathcal{X}}(t) = \tilde{A}_{\sigma(t)}\mathcal{X}(t)}.$$

3 Propagation through the switching intervals

Define the actual and nominal durations of interval i as

$$\Delta t_i = t_i - t_{i-1}, \quad \Delta \bar{t}_i = \bar{t}_i - \bar{t}_{i-1}.$$

Also define

$$F_i = \tilde{A}_{q_i} = \tilde{A}_{\bar{\sigma}(\bar{t}_{i-1})}.$$

The matrix F_i is constant throughout the i th interval.

For a constant matrix F_i , the solution over one interval is

$$\mathcal{X}(t_i) = e^{F_i \Delta t_i} \mathcal{X}(t_{i-1}).$$

For the first interval,

$$\mathcal{X}(t_1) = e^{F_1 \Delta t_1} \mathcal{X}(t_0).$$

For the second interval,

$$\begin{aligned} \mathcal{X}(t_2) &= e^{F_2 \Delta t_2} \mathcal{X}(t_1) \\ &= e^{F_2 \Delta t_2} e^{F_1 \Delta t_1} \mathcal{X}(t_0). \end{aligned}$$

Continuing interval by interval gives

$$\mathcal{X}(t_N) = e^{F_N \Delta t_N} e^{F_{N-1} \Delta t_{N-1}} \dots e^{F_2 \Delta t_2} e^{F_1 \Delta t_1} \mathcal{X}(t_0). \quad (1)$$

The order of these matrices is important: the transition associated with the first interval is the rightmost matrix.

The nominal state is propagated with the nominal interval durations:

$$\bar{\mathcal{X}}(t_N) = e^{F_N \Delta \bar{t}_N} e^{F_{N-1} \Delta \bar{t}_{N-1}} \dots e^{F_2 \Delta \bar{t}_2} e^{F_1 \Delta \bar{t}_1} \bar{\mathcal{X}}(t_0). \quad (2)$$

4 Exact error equation

Define the augmented-state error by

$$e(t) = \mathcal{X}(t) - \bar{\mathcal{X}}(t).$$

Let δt_i denote the perturbation of the i th interval duration:

$$\Delta t_i = \Delta \bar{t}_i + \delta t_i.$$

The initial state may also be perturbed, so that

$$\mathcal{X}(t_0) = \bar{\mathcal{X}}(t_0) + e(t_0).$$

Substitution in (1) gives

$$\begin{aligned} \mathcal{X}(t_N) &= e^{F_N(\Delta \bar{t}_N + \delta t_N)} e^{F_{N-1}(\Delta \bar{t}_{N-1} + \delta t_{N-1})} \dots \\ &\quad e^{F_2(\Delta \bar{t}_2 + \delta t_2)} e^{F_1(\Delta \bar{t}_1 + \delta t_1)} \left(\bar{\mathcal{X}}(t_0) + e(t_0) \right). \end{aligned} \quad (3)$$

Subtracting (2) from (3) produces the exact terminal error:

$$\begin{aligned}
e(t_N) = & e^{F_N(\Delta\bar{t}_N+\delta t_N)} e^{F_{N-1}(\Delta\bar{t}_{N-1}+\delta t_{N-1})} \dots \\
& e^{F_2(\Delta\bar{t}_2+\delta t_2)} e^{F_1(\Delta\bar{t}_1+\delta t_1)} \left(\bar{\mathcal{X}}(t_0) + e(t_0) \right) \\
& - e^{F_N\Delta\bar{t}_N} e^{F_{N-1}\Delta\bar{t}_{N-1}} \dots e^{F_2\Delta\bar{t}_2} e^{F_1\Delta\bar{t}_1} \bar{\mathcal{X}}(t_0).
\end{aligned} \tag{3}$$

5 First-order expansion of each transition matrix

For a constant matrix A ,

$$\frac{d}{ds} e^{As} = A e^{As} = e^{As} A.$$

The first-order Taylor expansion around $s = s_0$ is consequently

$$e^{A(s_0+\delta s)} = e^{As_0} + A e^{As_0} \delta s + \mathcal{O}(\delta s^2).$$

This identity explains the approximation used for every switching interval.

For clarity, define the nominal transition matrices individually:

$$\phi_1 = e^{F_1\Delta\bar{t}_1}, \quad \phi_2 = e^{F_2\Delta\bar{t}_2}, \quad \dots, \quad \phi_N = e^{F_N\Delta\bar{t}_N}.$$

Then each perturbed transition matrix has the approximation

$$\begin{aligned}
e^{F_1(\Delta\bar{t}_1+\delta t_1)} & \simeq \phi_1 + F_1\phi_1\delta t_1, \\
e^{F_2(\Delta\bar{t}_2+\delta t_2)} & \simeq \phi_2 + F_2\phi_2\delta t_2, \\
& \vdots \\
e^{F_N(\Delta\bar{t}_N+\delta t_N)} & \simeq \phi_N + F_N\phi_N\delta t_N.
\end{aligned}$$

Substituting these expressions into the exact error equation gives

$$\begin{aligned}
e(t_N) \simeq & (\phi_N + F_N\phi_N\delta t_N)(\phi_{N-1} + F_{N-1}\phi_{N-1}\delta t_{N-1}) \dots \\
& (\phi_2 + F_2\phi_2\delta t_2)(\phi_1 + F_1\phi_1\delta t_1) \left(\bar{\mathcal{X}}(t_0) + e(t_0) \right) \\
& - \phi_N\phi_{N-1} \dots \phi_2\phi_1 \bar{\mathcal{X}}(t_0).
\end{aligned} \tag{4}$$

6 Keeping only first-order terms

To see how the product is expanded, first consider three intervals. Up to first order,

$$\begin{aligned} & (\phi_3 + F_3\phi_3\delta t_3)(\phi_2 + F_2\phi_2\delta t_2)(\phi_1 + F_1\phi_1\delta t_1) \\ & \simeq \phi_3\phi_2\phi_1 + F_3\phi_3\phi_2\phi_1\delta t_3 \\ & \quad + \phi_3F_2\phi_2\phi_1\delta t_2 + \phi_3\phi_2F_1\phi_1\delta t_1. \end{aligned}$$

Terms such as

$$F_3\phi_3F_2\phi_2\phi_1\delta t_3\delta t_2$$

are discarded because they are second order.

For N intervals, the same expansion gives the nominal product

$$\phi_N\phi_{N-1}\cdots\phi_2\phi_1,$$

together with one first-order term for each duration perturbation:

$$\begin{aligned} & F_N\phi_N\phi_{N-1}\cdots\phi_2\phi_1\delta t_N, \\ & \phi_NF_{N-1}\phi_{N-1}\cdots\phi_2\phi_1\delta t_{N-1}, \\ & \phi_N\phi_{N-1}F_{N-2}\phi_{N-2}\cdots\phi_2\phi_1\delta t_{N-2}, \\ & \quad \vdots \\ & \phi_N\phi_{N-1}\cdots\phi_3F_2\phi_2\phi_1\delta t_2, \\ & \phi_N\phi_{N-1}\cdots\phi_3\phi_2F_1\phi_1\delta t_1. \end{aligned}$$

The matrices must remain in this order unless additional commutativity assumptions are imposed.

Multiplying by $\bar{\mathcal{X}}(t_0) + e(t_0)$ creates three kinds of terms:

1. the nominal-state term;
2. terms linear in the initial error $e(t_0)$;
3. terms linear in each duration perturbation δt_i .

The nominal-state term cancels with the final term in (4). Products of the form $\delta t_i e(t_0)$

are second order and are also discarded. Therefore,

$$\begin{aligned}
e(t_N) &\simeq \phi_N \phi_{N-1} \cdots \phi_2 \phi_1 e(t_0) \\
&\quad + \phi_N \phi_{N-1} \cdots \phi_3 \phi_2 F_1 \phi_1 \bar{\mathcal{X}}(t_0) \delta t_1 \\
&\quad + \phi_N \phi_{N-1} \cdots \phi_3 F_2 \phi_2 \phi_1 \bar{\mathcal{X}}(t_0) \delta t_2 \\
&\quad + \phi_N \phi_{N-1} \cdots F_3 \phi_3 \phi_2 \phi_1 \bar{\mathcal{X}}(t_0) \delta t_3 \\
&\quad \vdots \\
&\quad + \phi_N F_{N-1} \phi_{N-1} \cdots \phi_2 \phi_1 \bar{\mathcal{X}}(t_0) \delta t_{N-1} \\
&\quad + F_N \phi_N \phi_{N-1} \cdots \phi_2 \phi_1 \bar{\mathcal{X}}(t_0) \delta t_N.
\end{aligned} \tag{5}$$

7 Matrix form of the linearized model

Define the nominal state-transition matrix explicitly as

$$\Phi = \phi_N \phi_{N-1} \cdots \phi_2 \phi_1.$$

Collect the duration perturbations in the vector

$$\delta \mathbf{t} = \begin{bmatrix} \delta t_1 \\ \delta t_2 \\ \vdots \\ \delta t_{N-1} \\ \delta t_N \end{bmatrix}.$$

The sensitivity matrix is obtained by placing the coefficient of each δt_i in the corresponding column:

$$\begin{aligned}
\Gamma = &\begin{bmatrix} \phi_N \phi_{N-1} \cdots \phi_3 \phi_2 F_1 \phi_1 \bar{\mathcal{X}}(t_0), \\ \phi_N \phi_{N-1} \cdots \phi_3 F_2 \phi_2 \phi_1 \bar{\mathcal{X}}(t_0), \\ \phi_N \phi_{N-1} \cdots F_3 \phi_3 \phi_2 \phi_1 \bar{\mathcal{X}}(t_0), \\ \vdots, \\ \phi_N F_{N-1} \phi_{N-1} \cdots \phi_2 \phi_1 \bar{\mathcal{X}}(t_0), \\ F_N \phi_N \phi_{N-1} \cdots \phi_2 \phi_1 \bar{\mathcal{X}}(t_0) \end{bmatrix}.
\end{aligned}$$

Equation (5) can now be written compactly as

$$\boxed{e(t_N) \simeq \Phi e(t_0) + \Gamma \delta \mathbf{t}}. \quad (6)$$

This is the first-order terminal-error model. It separates the effect of the initial-state error from the effect of perturbations in the switching interval durations.